Problem 11 | Problems T6andT7_Electric and Magnetic Properties.pdf #11 | Soft and hard magnets · B-H loop

Hysteresis Loops of Soft and Hard Magnetic Materials

Original statement · Problems T6andT7_Electric and Magnetic Properties.pdf, p. 3

11. Draw the hysteresis curve of a soft magnetic material and a hard magnetic material and indicated in both cases the most significant features. What are the main differences between these two materials?

The sheet prints no solution for this problem.

HYPOTHESES & PARAMETERS

Hypotheses

1. **Saturated loop, quasi-static.** The loop is the one obtained by cycling the field slowly between $+H_{max}$ and $-H_{max}$, with H_{max} large enough to reach saturation. The drawing is a schematic, as the statement asks for a sketch and gives no numbers.
2. **Features to indicate.** The features of the slides are the saturation induction B_s (saturation magnetization M_s), the remanence B_r, the coercivity H_c and the permeabilities μ_i and μ_{max} [Session 15 Slides 16-17].
3. **Soft and hard.** The two classes are separated by the coercivity, $H_c = 10\,000$ A/m [Session 15 Slide 18].
4. **Illustration with data.** The numbers come from the tables of Slide 24 of Session 15: Table 20-4 (soft materials, with H_c in A/m) and Table 20-6 (hard materials, with $\mu_0 H_c$ in tesla). $\mu_0 = 1.257 \times 10^{-6}$ H/m [Session 15 Slide 4].

Data

Soft material	$\mu_{r,max}$	H_c in A/m	B_r in T
Ingot iron (99.8 % Fe)	5000	80	0.77
Fe-3 % Si, grain-oriented	50 000	7	1.20
Superalloy (Fe-5 % Mo-80 % Ni)	450 000	0.4	not printed

Hard material	$\mu_0 M_r$ in T	$\mu_0 H_c$ in T
Co-steel (Fe-Co)	1.07	0.02
Alnico-5 (Fe-Co-Al-Ni)	1.05	0.06
Ferrite ($BaFe_{12}O_{19}$)	0.42	0.31
Sm-Co ($SmCo_5$)	0.87	0.80

Hard material	$\mu_0 M_r$ in T	$\mu_0 H_c$ in T
Nd-Fe-B ($Nd_2Fe_{14}B$)	1.23	1.21

Unknown: the two loops with their features, and the differences between the materials.

MODEL & STRATEGY

The loop comes from the motion of the Bloch walls. In a soft material the walls move easily, so a small field is enough to magnetize the material and to demagnetize it again. In a hard material the walls are difficult to move, so a large field is needed in both directions [Session 15 Slide 18].

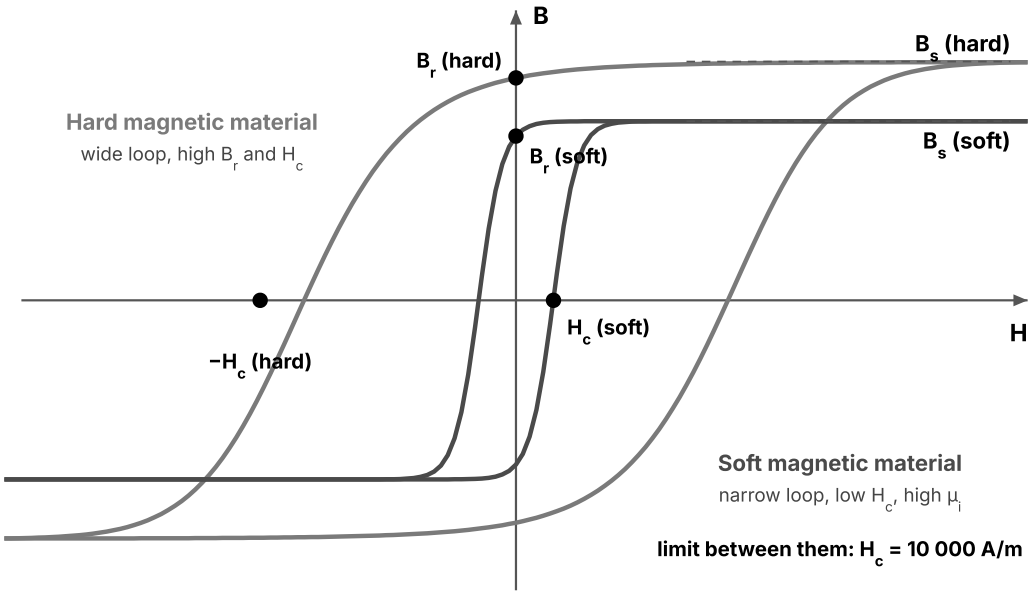


Figure 1. B-H hysteresis loops: a soft material has a narrow loop (low coercivity Hc, high initial permeability), a hard material a wide one (high remanence Br, saturation Bs and Hc); the slide fixes the limit at Hc = 10 000 A/m (schematic).

The permeabilities are read on the initial magnetization curve, which starts from the demagnetized state before the loop is traced:

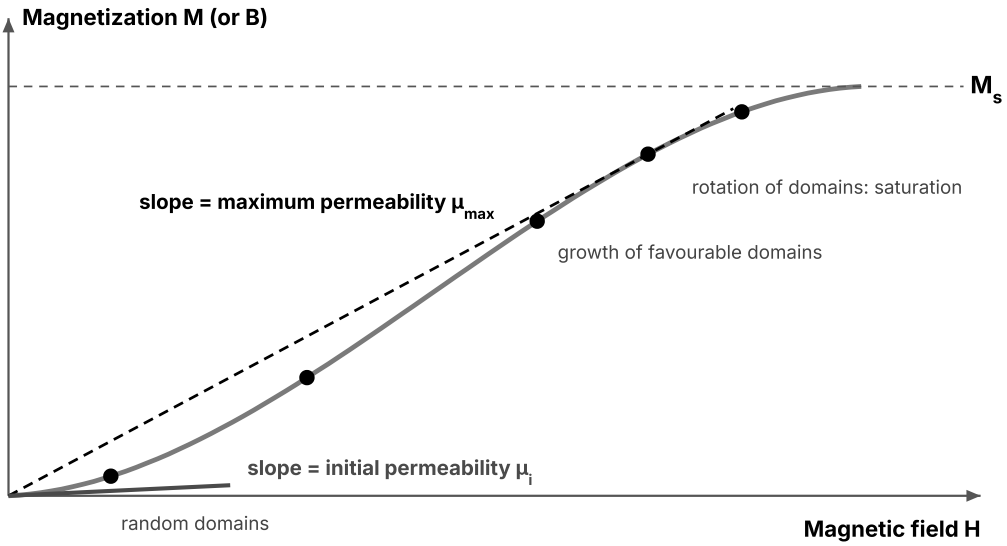


Figure 2. Magnetization against field starting from the demagnetized state: slow growth (initial permeability), fast growth of favourable domains (maximum permeability), rotation of domains and saturation Ms (schematic).

Strategy.

1. Follow the loop step by step and mark where each feature appears.
2. Put numbers on the features with the tables of Slide 24, converting the hard-material coercivity to A/m so that both classes are compared with the same unit and with the limit of 10 000 A/m.
3. Compare the two classes feature by feature.

STEP-BY-STEP CALCULATION

1. Path of the loop [Session 15 Slides 16-17]

- Starting from the demagnetized state ($H = 0$, domains at random) the initial curve rises to saturation: $B = B_s$.
- Reducing H to zero does not bring B back to zero: the induction left is the **remanence** B_r.
- A field in the opposite direction, of magnitude H_c, brings B to zero: the **coercivity**.
- Going on to $-H_{max}$ gives $-B_s$, and the return branch is the mirror image through the origin.

The slope of the initial curve is the initial permeability μ_i, and the line from the origin tangent to the curve gives the maximum permeability μ_{max}.

2. Features on each loop

Feature	Where	Soft material	Hard material
μ_i, μ_{max}	slope of the initial curve	high	low μ_i
H_c	crossing of the H axis	low, below 10 000 A/m	high, above 10 000 A/m
B_r	crossing of the B axis	printed as a feature, not as a rule	high, printed on Slide 18

Feature	Where	Soft material	Hard material
B_s	plateau	reached with a small field	reached only with a large field
Loop area	enclosed by the loop	small	large

3. Hard-material coercivity in A/m

From $\mu_0 H_c$ in tesla, with $\mu_0 = 1.257 \times 10^{-6}$ H/m:

$$H_c = \frac{\mu_0 H_c}{\mu_0}$$

For Nd-Fe-B:

$$H_c = \frac{1.21\ \mathrm{T}}{1.257 \times 10^{-6}\ \mathrm{H/m}} = 9.626 \times 10^{5}\ \mathrm{A/m}$$

The same division for the others:

Hard material	H_c in A/m	Above 10 000 A/m?
Co-steel	1.59×10^{4}	yes
Alnico-5	4.77×10^{4}	yes
Ferrite	2.47×10^{5}	yes
Sm-Co	6.36×10^{5}	yes
Nd-Fe-B	9.63×10^{5}	yes

Ratio of the strongest hard material to the soft ones of the table:

$$\frac{9.626 \times 10^{5}\ \mathrm{A/m}}{80\ \mathrm{A/m}} = 1.2 \times 10^{4} \qquad \frac{9.626 \times 10^{5}\ \mathrm{A/m}}{7\ \mathrm{A/m}} = 1.4 \times 10^{5}$$

4. Energy lost per cycle (extension, order of magnitude)

The energy dissipated per unit volume in one cycle is the area enclosed by the loop, $W = \oint H\, dB$ (this relation is not printed on the slides). For an idealized rectangular loop of width $2H_c$ and height $2B_r$ the area is

$$W \approx 2H_c \times 2B_r = 4\,H_c\,B_r$$

For the grain-oriented Fe-3 % Si and for Nd-Fe-B:

$$W_{soft} \approx 4 \times 7\ \mathrm{A/m} \times 1.20\ \mathrm{T} = 33.6\ \mathrm{J/m^3} \qquad W_{hard} \approx 4 \times 9.626 \times 10^5\ \mathrm{A/m} \times 1.23\ \mathrm{T} = 4.74 \times 10^6\ \mathrm{J/m^3}$$

The ratio is about 1.4×10^5, the same as the ratio of coercivities, because the two remanences are similar. This is why the soft materials are the ones used under alternating fields, where the cycle repeats many times per second, and the hard ones are used as permanent magnets, where it never repeats [Session 15 Slides 18-19].

5. Main differences

	Soft magnetic material	Hard magnetic material
Bloch walls	easy to move	difficult to move
Loop	narrow	wide
Coercivity	$H_c < 10\,000$ A/m	$H_c > 10\,000$ A/m
Permeability	high initial permeability	low initial permeability
Losses per cycle	low	high
Demagnetization	easy	resistant
Use	cores of transformers, generators and motors; materials under alternating fields	permanent magnets

RESULT & CHECKS

Checks

- **Units of H_c.** $[\mu_0 H_c]/[\mu_0] = \mathrm{T/(H/m)} = (\mathrm{Wb/m^2})/(\mathrm{Wb/(A \cdot m)}) = \mathrm{A/m}$.
- **Units of the energy.** $[H\,B] = (\mathrm{A/m})(\mathrm{T}) = \mathrm{A \cdot T/m}$, and with $\mathrm{T} = \mathrm{N/(A \cdot m)}$ this is $\mathrm{N/m^2} = \mathrm{J/m^3}$, an energy per unit volume.
- **The limit of the slide.** All the soft materials of the table are below 10 000 A/m (the highest is 100 A/m for the low-carbon steel of Table 20-4) and all the hard ones above it. Co-steel, at 1.59×10^4 A/m, is the closest to the limit.

- **Limit $H_c \to 0$.** The loop collapses into a single line with no area and no loss, the ideal soft material. For a very large H_c the loop is so wide that the material keeps its magnetization against the fields it meets in use, the ideal permanent magnet.

Interpretation

The decisive difference is the coercivity, and everything else follows from the mobility of the Bloch walls: the width of the loop, the losses and the use. The remanence alone does not separate the classes. The grain-oriented Fe-3 % Si has $B_r = 1.20$ T, more than the ferrite magnet (0.42 T) and about the same as Nd-Fe-B (1.23 T). Slide 18 lists a high remanence among the properties of the hard materials; in the data of Slide 24 it is the coercivity that is orders of magnitude apart (7 to 100 A/m against 10^4 to 10^6 A/m), not the remanence.

Check against the official solution

Item	Official	Computed	Comment
Loops and features	not printed on the sheet	B_s, B_r and H_c marked on both loops; μ_i and μ_{max} on the initial magnetization curve	consistent with Session 15 Slides 16-18
Differences	not printed on the sheet	coercivity, loop width, permeability, losses and use	consistent with Session 15 Slides 18-19; the energy per cycle and the numbers of Slide 24 are an extension